

File Handling in Spring Boot – Task PDF

1. File Upload (MultipartFile) – Theory

File upload allows clients to send files to the server using multipart/form-data. Spring Boot provides MultipartFile to handle uploaded files. It supports validation, size limits, and type checking.

Practical Tasks

- 1 Create REST API to upload file using MultipartFile
- 2 Validate file type and size
- 3 Store file on server
- 4 Return file metadata as response

2. File Download – Theory

File download enables users to retrieve stored files from the server. Spring Boot uses ResponseEntity and Resource to stream files efficiently.

Practical Tasks

- 1 Create API to download file by ID or name
- 2 Set proper headers for file download
- 3 Handle file-not-found scenarios

3. Image Handling – Theory

Image handling includes upload, validation, storage, and retrieval of image files. Images can be stored in file system or database as BLOBs.

Practical Tasks

- 1 Upload image file using MultipartFile
- 2 Validate image format and size
- 3 Display image in browser using REST API

4. PDF Generation – Theory

PDF generation is used for reports, invoices, and tickets. PDFs can be dynamically created using Java libraries and streamed to users.

Practical Tasks

- 1 Generate PDF dynamically from data
- 2 Download generated PDF via API
- 3 Store generated PDF on server

5. Storing Files – DB vs File System

Files can be stored either in database as BLOBs or in file system with metadata in DB. File system storage is preferred for large files, while DB storage is suitable for small files.

Mini Project Task

Build a document management module supporting file upload, download, image display, PDF generation, and metadata storage.